Cambridge (CIE) A Level Maths: Pure 3

Your notes

Differentiation of Parametric Equations

Contents

- Basics of Parametric Equations
- Eliminating the Parameter of Parametric Equations
- Sketching Graphs of Parametric Equations
- Parametric Differentiation

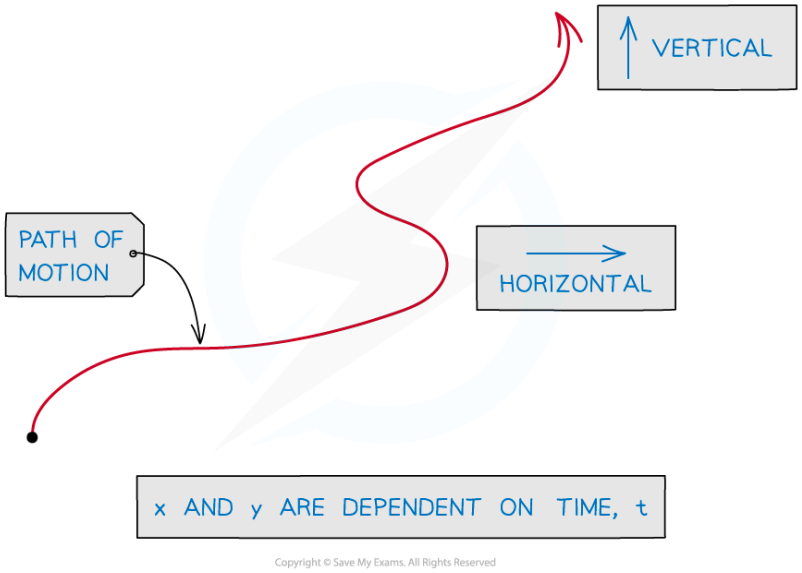
© 2025 Save My Exams, Ltd.

Your notes

Basics of parametric equations

What are parametric equations

- Graphs are usually described by a **Cartesian equation**
  - The equation involves **x** and **y** only
- Equations like this can sometimes be rearranged into the form, **y = f(x)**



- In **parametric equations** both **x** and **y** are dependent on a third variable
  - This is called a **parameter**
  - **t** and **θ** are often used as parameters
- A common example ...
  - **x** is the horizontal position of an object
  - **y** is the vertical position of an object
  - and the position of the object is dependent on time **t**
- **x** is a function of **t**, **y** is a function of **t**
  - **x = f(t)**
  - **y = g(t)**

How do I plot parametric equations?

- It is still possible to plot a graph of **y** against **x** from their parametric equations

Your notes

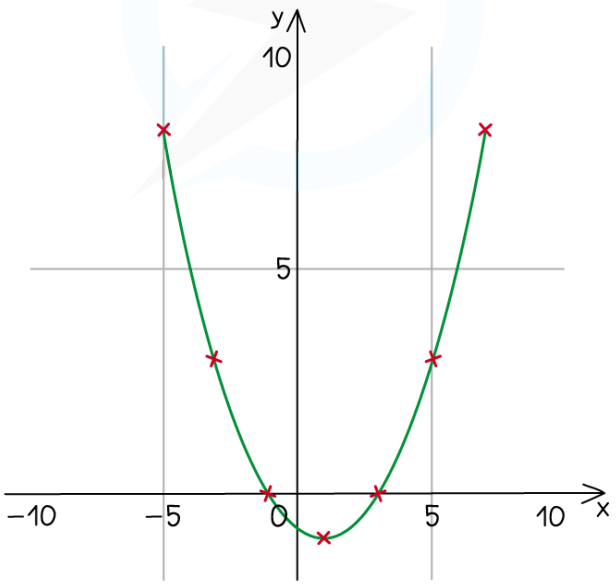
PLOTTING A GRAPH FROM PARAMETRIC EQUATIONS

e.g. PLOT THE GRAPH GIVEN BY THE PARAMETRIC EQUATIONS $x = 2t + 1$ AND $y = t^2 - 1$ FOR $-3 \leqslant t \leqslant 3$.

CONSTRUCT A TABLE OF VALUES

t	−3	−2	−1	0	1	2	3
x	−5	−3	−1	1	3	5	7
y	8	3	0	−1	0	3	8

PLOT THE GRAPH OF y AGAINST x

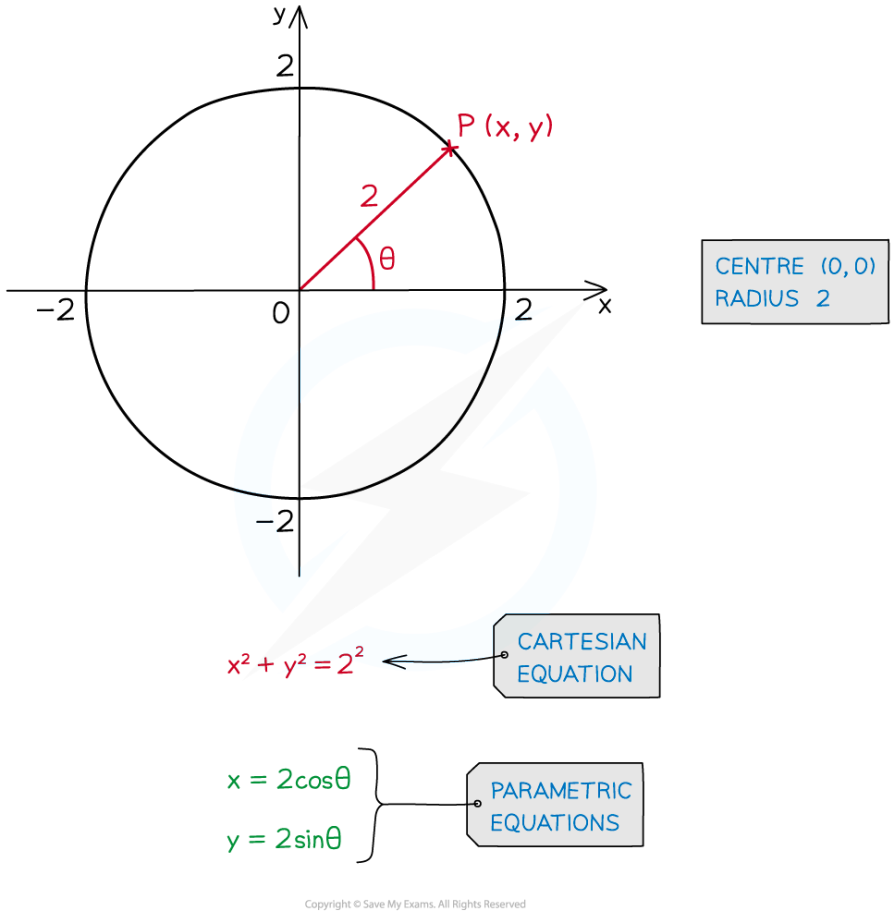


Copyright © Save My Exams. All Rights Reserved

- Also see Parametric Equations – Sketching Graphs

What are the parametric equations of a circle?

Your notes



- For a circle, centre (0, 0) and radius **r**
  - **x = rcos θ**
  - **y = rsin θ**
  - (Note that **r** is constant, this is not two parameters)
- For a circle, centre (**a**, **b**) and radius **r**
  - **x = rcos θ + a**
  - **y = rsin θ + b**

Worked Example

Your notes

(a) Write down the radius and the centre of the circle defined by the parametric equations

$$x = 3\cos\theta$$

and

$$y = 3\sin\theta - 4$$

(b) Hence write down the Cartesian equation of the circle

a) CENTRE: $(0, -4)$
RADIUS: 3

$x = r\cos\theta + a$
$y = r\sin\theta + b$

b) $x^2 + (y+4)^2 = 9$

$r^2 = 3^2 = 9$

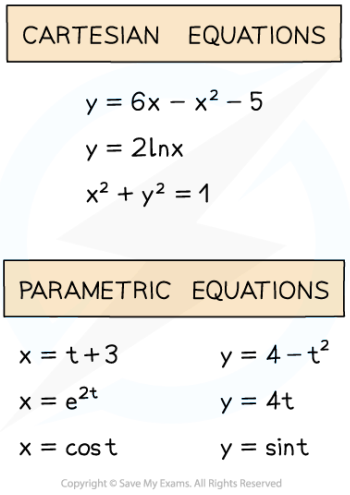
CENTRE $(0, -4)$

Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

Eliminating the parameter of parametric equations

What does eliminating the parameter mean?



- In parametric equations, **x = f(t)** and **y = g(t)**
- There is still a connection directly linking **x** and **y**
  - This will be the **Cartesian** equation of the graph

How do I find the Cartesian equation from parametric equations?

Your notes

e.g. FIND A CARTESIAN EQUATION FOR THE CURVE GIVEN PARAMETRICALLY AS

$$x = \frac{t-3}{4} \qquad y = e^{2t}$$

STEP 1 REARRANGE ONE EQUATION TO MAKE t THE SUBJECT

$4x = t - 3$

$t = 4x + 3$

BOTH EQUATIONS EASY TO REARRANGE

STEP 2 SUBSTITUTE INTO OTHER EQUATION

$y = e^{2(4x+3)}$

STEP 3 REARRANGE INTO CARTESIAN FORM

$y = e^{8x+6}$

NOT MUCH TO DO HERE!

Copyright © Save My Exams. All Rights Reserved

- STEP 1: Rearrange one of the equations to make t the subject
  - Either **t = p(x)** or **t = q(y)**
- STEP 2: Substitute into the other equation
- STEP 3 Rearrange into the desired (Cartesian) form

How can I eliminate the parametric using trigonometric identities?

Your notes

e.g. FIND A CARTESIAN EQUATION FOR THE CURVE GIVEN PARAMETRICALLY AS

$$x = 3 + \cos t \qquad y = 2 - \sin t$$

STEP 1 REARRANGE BOTH EQUATIONS INTO THE FORM $\cos t = \ldots$ AND $\sin t = \ldots$

$$\cos t = x - 3$$

$$\sin t = 2 - y$$

STEP 2 SQUARE BOTH SIDES OF BOTH EQUATIONS

$$\cos^2 t = (x-3)^2$$

$$\sin^2 t = (2-y)^2$$

STEP 3 ADD THE EQUATIONS TOGETHER

$$\cos^2 t + \sin^2 t = (x-3)^2 + (2-y)^2$$

STEP 4 USE THE TRIG IDENTITY "$\sin^2 x + \cos^2 x \equiv 1$" TO ELIMINATE t

$$1 = (x-3)^2 + (2-y)^2$$

STEP 5 REARRANGED INTO DESIRED FORM

$$(x-3)^2 + (y-2)^2 = 1$$

EQUATION OF A CIRCLE CENTRE: (3, 2) RADIUS: 1

$(y-2)^2 = (2-y)^2$

Copyright © Save My Exams. All Rights Reserved

- STEP 1 Rearrange both equations into the forms "cos t = ..." and "sin t = ..."
- STEP 2 Square BOTH sides of BOTH equations
- STEP 3 Add the equations together
- STEP 4 The trig identity "$\sin^2 x + \cos^2 x \equiv 1$" eliminates **t**
- STEP 5 Rearrange into desired (Cartesian) form
  - This technique is seen in Trigonometric Identities

Your notes

Examiner Tips and Tricks

When choosing which equation to rearrange, aim for "as simple as possible":

- Linear equations are simpler than quadratics
  - eg Rearrange $x = 2t + 3$

    or

    $y = 3t^2 + 3t - 4$?
- Single exponential terms are quite easy to deal with
  - eg $x = e^t \rightarrow t = \ln x$

Trig identities may be needed and remember **squared** terms are good!

- eg $\sin^2 x + \cos^2 x \equiv 1$

Worked Example

Your notes

Find the Cartesian equation for the curve C defined by the parametric equations

$$x = 3\sin 2t$$

and

$$y = 2\cos 2t$$

- THIS IS A sin/cos BASED QUESTION
- AIM FOR "SQUARE AND ADD"

STEP 1

REARRANGE BOTH INTO FORM "sin t = ..." AND "cos t = ..."

$\frac{x}{3} = \sin 2t$ $\quad$ $\frac{y}{2} = \cos 2t$

STEP 2 & 3

"SQUARE AND ADD"

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = \sin^2 2t + \cos^2 2t$$

STEP 4

ELIMINATE t

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$

STEP 5

REARRANGE INTO DESIRED FORM

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

EQUATION OF AN ELIPSE

$$4x^2 + 9y^2 = 36$$

Copyright © Save My Exams. All Rights Reserved
save my exams

Your notes

Sketching graphs of parametric equations

How do I sketch a graph from parametric equations?

- **Plotting** a graph is covered in Parametric Equations - Basics

e.g. SKETCH THE GRAPH OF y AGAINST x FOR THE PARAMETRIC EQUATIONS

$$x = 3t + 1 \qquad y = t^2 - 4$$

$3t + 1 = 0$

$t = -\frac{1}{3} \longrightarrow y = (-\frac{1}{3})^2 - 4$

$\left(0, -\frac{35}{9}\right) \qquad y = -\frac{35}{9}$

y–AXIS INTERCEPT
$x = 0$

$t^2 - 4 = 0$

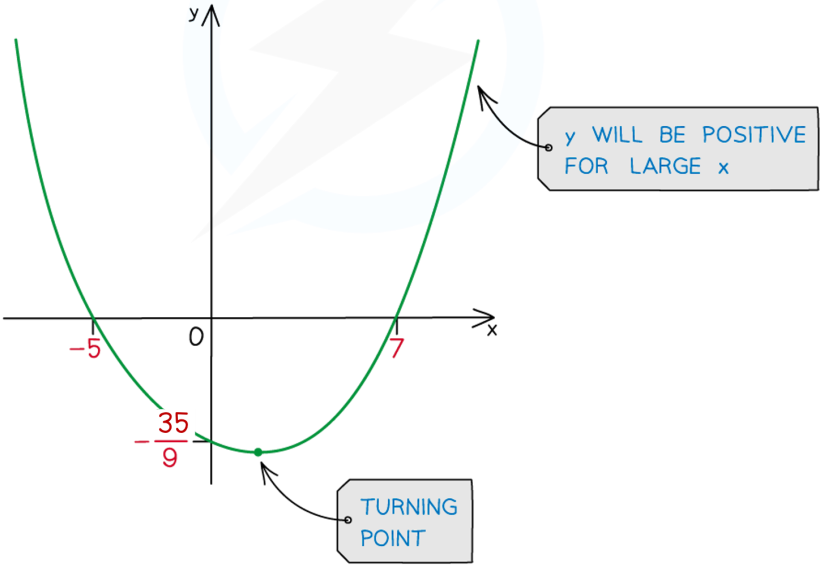
$t = \pm 2 \longrightarrow x = 3 \times 2 + 1 = 7$

$x = 3 \times -2 + 1 = -5$

$(-5, 0)$

$(7, 0)$

x–AXIS INTERCEPT(S)
$y = 0$

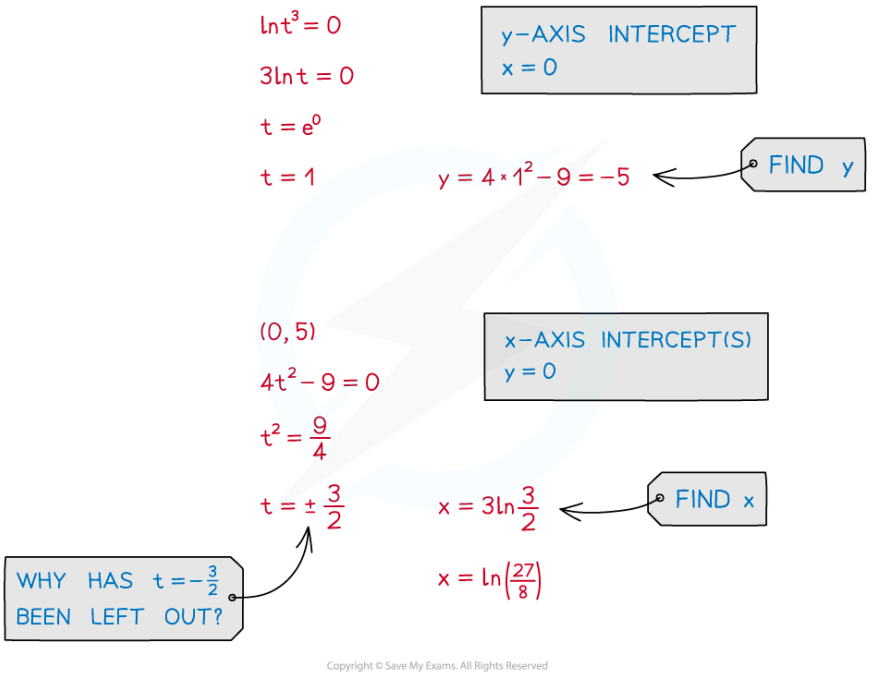


IT IS NOT REQUIRED BUT THE CARTESIAN EQUATION IS $9y = x^2 - 2x - 35$ WHICH IS A QUADRATIC

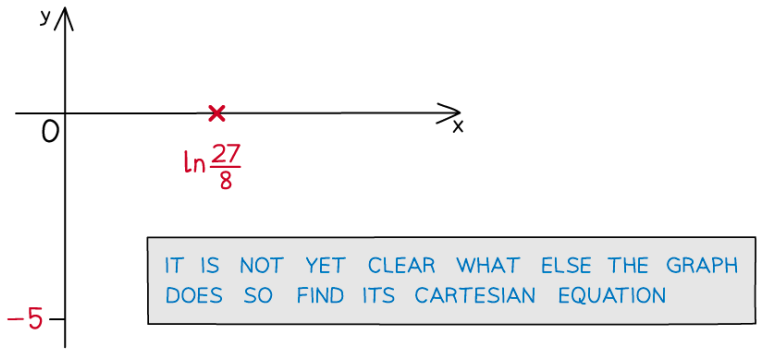
Copyright © Save My Exams. All Rights Reserved

- Still find the key features of a graph ...

- … the **y**-axis intercept
  - … the **x**-axis intercept(s)
  - … asymptotes
  - … location of (and if required coordinates of) stationary points (see Parametric Differentiation)
- **Sketch** these points and join up accordingly



Your notes



IT IS NOT YET CLEAR WHAT ELSE THE GRAPH DOES SO FIND ITS CARTESIAN EQUATION

$x = \ln t^3$

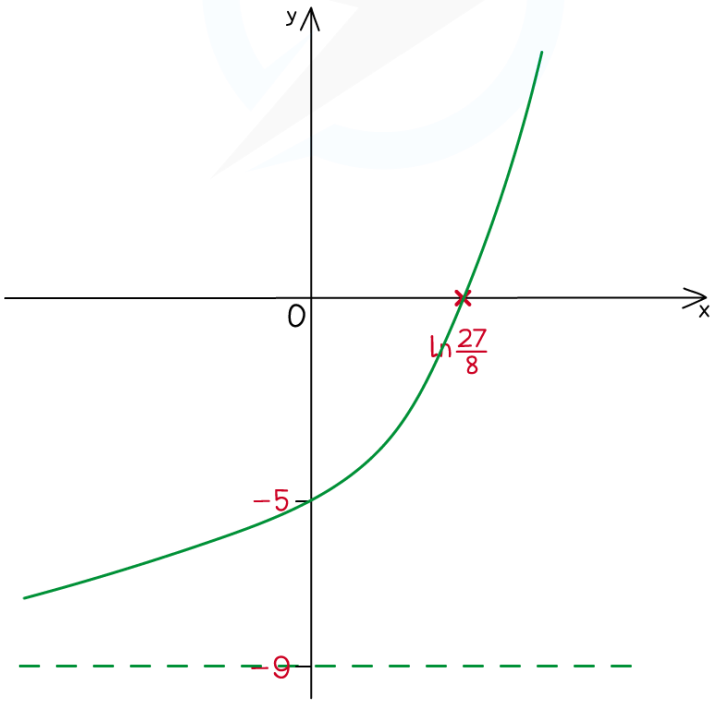
$x = 3\ln t$

$\ln t = \frac{x}{3}$

$t = e^{\frac{x}{3}}$ $\qquad y = 4\left(e^{\frac{x}{3}}\right)^2 - 9$

$y = 4e^{\frac{2}{3}x} - 9$

A POSITIVE EXPONENTIAL GRAPH
∘ ASYMPTOTE AT $y = -9$



Copyright © Save My Exams. All Rights Reserved

Your notes

Examiner Tips and Tricks

- Not all curves defined parametrically lead to familiar shaped graphs and it may be worth plotting a few extra points by calculating them.
- Your calculator may be able to produce a table of values quickly, however, remember you are sketching and not plotting.
- It may be easier to find the Cartesian equation first and draw the graph from that – this will depend on the question.
- It is only a **definite** strategy if you cannot make progress otherwise.
- If you are given the sketch of a graph it is usually only for reference and can be used to check answers for axes intercepts, etc.

Your notes

Worked Example

Your notes

Sketch the graph of the curve defined by the parametric equations

$$x = e^t \text{ and } y = t^2 - t$$

stating clearly the coordinates of any points where the curve intercepts the coordinate axes.

DEDUCE $x > 0$ SO y-AXIS IS AN ASYMPTOTE

$x = e^t = 0$
NO SOLUTIONS

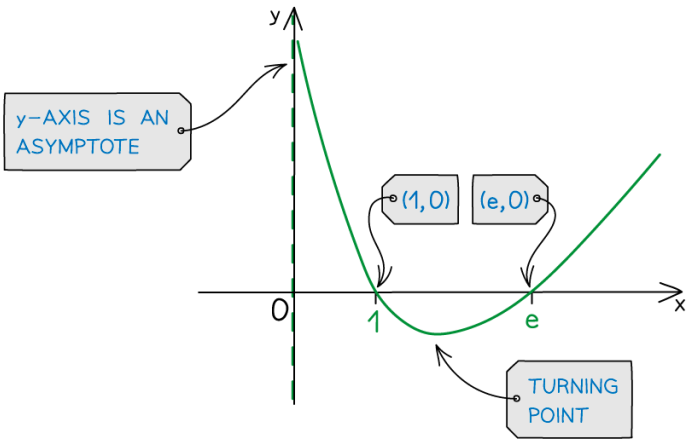
y-AXIS INTERCEPT
$x = 0$

$y = t^2 - t = 0$
$t(t-1) = 0$
$t = 0, \quad t = 1$

x-AXIS INTERCEPT(S)
$y = 0$

AT $t = 0, \quad x = e^0 = 1$
AT $t = 1, \quad x = e^1 = e$

THERE WILL BE A TURNING POINT BETWEEN $x = 1$ AND $x = e$



WHY IS THE GRAPH THIS WAY UP?

ANSWER: AS $t \to -\infty$, $x \to 0$, $y \to \infty$

Your notes

Parametric differentiation

How do I find dy/dx from parametric equations?

- Ensure you are familiar with Parametric Equations – Basics first

PARAMETRIC DIFFERENTIATION

$x = f(t)$ $\quad$ $y = g(t)$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

WHERE $\frac{dt}{dx} = \frac{1}{\frac{dx}{dt}}$

NOTE 1: $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ IS EQUIVALENT TO ABOVE

NOTE 2: $\frac{dy}{dx}$ WILL BE A FUNCTION OF t

Copyright © Save My Exams. All Rights Reserved

- This method uses the chain rule and the reciprocal property of derivatives
  - $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$
  - $\frac{dt}{dx} = 1 \div \frac{dx}{dt}$
- Equivalently, $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$
- $\frac{dy}{dx}$ will be in terms of t – this is fine
  - Questions usually involve finding gradients, tangents and normals

- The chain rule is always needed when there are three variables or more – see Connected Rates of Change

Your notes

How do I find gradients, tangents and normals from parametric equations?

- To find a gradient …
  - STEP 1: Find **dx/dt** and **dy/dt**
  - STEP 2: Find **dy/dx** in terms of **t**

  Using either **dy/dx = dy/dt ÷ dx/dt**

  or **dy/dx = dy/dt × dt/dx** where **dt/dx = 1 ÷ dx/dt**

  - STEP 3: Find the value of **t** at the required point
  - STEP 4: Substitute this value of **t** into **dy/dx** to find the gradient

© 2025 Save My Exams, Ltd.

Your notes

e.g. FIND THE GRADIENT OF THE CURVE DEFINED PARAMETRICALLY BY

$$x = 2t - 3$$
$$y = t^3 - 4t + 1$$

AT THE POINT WHERE $x = 5$

STEP 1 FIND $\frac{dx}{dt}$ AND $\frac{dy}{dt}$

$\frac{dx}{dt} = 2$ $\frac{dy}{dt} = 3t^2 - 4$

STEP 2 FIND $\frac{dy}{dt}$ (IN TERMS OF t)

$$\frac{dy}{dx} = (3t^2 - 4) \times \frac{1}{2}$$

$\frac{dy}{dt}$ $\frac{1}{\frac{dy}{dt}}$

$$\frac{dy}{dx} = \frac{1}{2}(3t^2 - 4)$$

STEP 3 FIND THE VALUE OF t AT THE REQUIRED POINT

AT $x = 5$, $5 = 2t - 3$

$t = 4$

STEP 4 SUBSTITUTE THIS VALUE OF t INTO $\frac{dy}{dx}$ TO FIND THE GRADIENT

AT $x = 5$, $\frac{dy}{dx} = \frac{1}{2}(3 \times 4^2 - 4)$

$t = 4$,

$$\frac{dy}{dx} = 22$$

Copyright © Save My Exams. All Rights Reserved

- to then go on to find the equation of a tangent ...
  - STEP 5: Find the **y** coordinate
  - STEP 6: Use the gradient and point to find the equation of the tangent

Your notes

e.g. FIND THE EQUATION OF THE TANGENT TO THE CURVE DEFINED PARAMETRICALLY BY

$$x = 2t - 3$$
$$y = t^3 - 4t + 1$$

AT THE POINT WHERE $x = 5$

TANGENT: "$y - y_1 = m(x - x_1)$"

$m = \frac{dy}{dx}$ AT $x = 5$

FROM STEP 4 → $m = 22$

STEP 5 FIND THE y–COORDINATE

TO FIND THE EQUATION OF A LINE WE NEED THE GRADIENT, m, AND A POINT (x_1, y_1) THE LINE PASSES THROUGH

AT $x = 5$, $t = 4$, $y = 4^3 - 4 \times 4 + 1 = 49$

FROM STEP 3 → $t = 4$

STEP 6 USE GRADIENT AND POINT TO FIND EQUATION OF TANGENT

$y - 49 = 22(x - 5)$

$y = 22x - 61$

Copyright © Save My Exams. All Rights Reserved

- To find a normal...
  - STEP 7: Use perpendicular lines property to find the gradient of the normal $\mathbf{m_1 \times m_2 = -1}$
  - STEP 8: Use gradient and point to find the equation of the normal $\mathbf{y - y_1 = m(x - x_1)}$

How do I solve problems using parametric differentiation?

- Questions may require use of tangents and normals as per the coordinate geometry sections
  - Find points of intersection between a tangent/normal and **x/y** axes
  - Find areas of basic shapes enclosed by axes and/or tangents/normal
- Find stationary points (**dy/dx = 0**)

Your notes

e.g. FIND THE STATIONARY POINTS ON THE CURVE DEFINED PARAMETRICALLY BY

$$x = 2t - 3$$
$$y = t^3 - 4t + 1$$

$$\frac{dy}{dx} = \frac{1}{2}(3t^2 - 4)$$

FROM EARLIER EXAMPLES

$$\frac{1}{2}(3t^2 - 4) = 0$$

$\frac{dy}{dx} = 0$ AT STARTING POINTS

$$t^2 = \frac{4}{3}$$

$$t = \pm\frac{2}{\sqrt{3}}$$

SAME AS $t = \pm\frac{2\sqrt{3}}{3}$

WORK OUT x & y FROM t

AT $t = \frac{2}{\sqrt{3}}$, $x = \frac{-9 + 4\sqrt{3}}{3}$ $y = \frac{9 - 16\sqrt{3}}{9}$

AT $t = \frac{-2}{\sqrt{3}}$, $x = \frac{-9 - 4\sqrt{3}}{3}$ $y = \frac{9 + 16\sqrt{3}}{9}$

CAREFUL WITH CALCULATION DISPLAY: $-\frac{9 + 4\sqrt{3}}{3}$

∴ THE COORDINATES OF THE STATIONARY POINTS ARE

$$\left(\frac{-9 + 4\sqrt{3}}{3}, \frac{9 - 16\sqrt{3}}{9}\right)$$

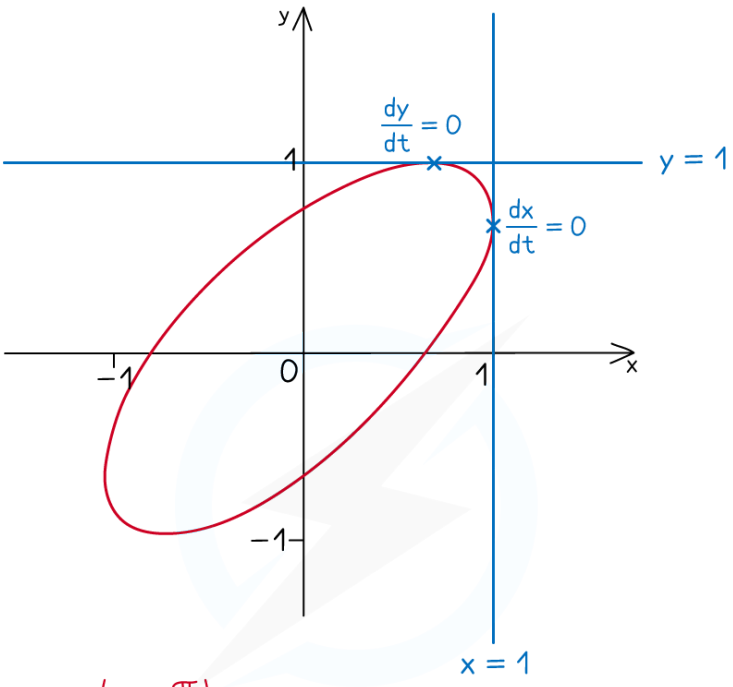
AND

$$\left(\frac{-9 - 4\sqrt{3}}{3}, \frac{9 + 16\sqrt{3}}{9}\right)$$

Copyright © Save My Exams. All Rights Reserved

- You may also be asked about horizontal and vertical tangents
  - At horizontal (parallel to the **x**-axis) tangents, **dy/dt = 0**
  - At vertical (parallel to **y**-axis) tangents, **dx/dt = 0**

Your notes



$x = \cos\left(2t - \frac{\pi}{4}\right)$

$y = \sin\left(2t + \frac{\pi}{2}\right)$

- $x = -1$ IS ALSO A VERTICAL TANGENT
- $y = -1$ IS ALSO A HORIZONTAL TANGENT

Copyright © Save My Exams. All Rights Reserved

Worked Example

Your notes

Curve C is defined parametrically by the equations

$$x = 2\cos 3t \text{ and } y = \sin(3t + \frac{\pi}{3})$$

where $-\frac{\pi}{3} \leq t \leq 0$

The line l_1 is the normal to the curve at the point where $x = 1$.

(a) Find the equation of l_1

(b) Find the exact area of the triangle made by l_1 and the coordinate axes

a) l_1: "$y - y_1 = m(x - x_1)$"

STEP 1 FIND $\frac{dx}{dt}$ AND $\frac{dy}{dt}$

$$\frac{dx}{dt} = -6\sin 3t$$

$$\frac{dy}{dt} = 3\cos\left(3t + \frac{\pi}{3}\right)$$

BOTH INVOLVE CHAIN RULE

STEP 2 FIND $\frac{dy}{dx}$ (IN TERMS OF t)

$$\frac{dy}{dx} = \frac{3\cos\left(3t + \frac{\pi}{3}\right)}{-6\sin 3t}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

$$= \frac{-\cos\left(3t + \frac{\pi}{3}\right)}{2\sin 3t}$$

STEP 3 FIND THE VALUE OF t AT $x = 1$

$$2\cos 3t = 1$$

$$\cos 3t = \frac{1}{2}$$

$$3t = -\frac{\pi}{3}$$

$$t = -\frac{\pi}{9}$$

EXTREME CARE

$-\frac{\pi}{3} \leqslant t \leqslant 0$

$\therefore -\pi \leqslant 3t \leqslant 0$

STEP 4 FIND $\frac{dy}{dx}$ AT $t = -\frac{\pi}{9}$

Your notes

$$\frac{dy}{dx} = \frac{-\cos(0)}{2\sin\left(-\frac{\pi}{3}\right)} = \frac{\sqrt{3}}{3}$$

STEP 5 FIND THE y–COORDINATE USING THE VALUE OF t

$$y = \sin\left(3 \times \left(-\frac{\pi}{9}\right) + \frac{\pi}{3}\right)$$

$$y = \sin(0) = 0$$

STEP 6 "$m_1 \times m_2 = -1$"

GRADIENT OF NORMAL, $m = \frac{-1}{\frac{\sqrt{3}}{3}} = -\sqrt{3}$

STEP 7 "$y - y_1 = m(x - x_1)$"

$$y - 0 = -\sqrt{3}(x - 1)$$

$$l_1 : y = -\sqrt{3}x + \sqrt{3}$$

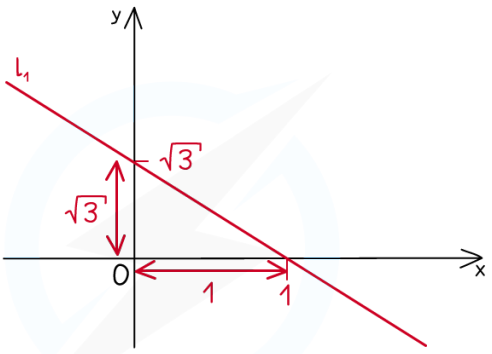
Copyright © Save My Exams. All Rights Reserved
save my exams

b) $y = -\sqrt{3}x + \sqrt{3}$

FIND WHERE LINE INTERCEPTS AXES

AT $x = 0$, $y = \sqrt{3}$

AT $y = 0$, $x = 1$



A SKETCH MAKES THE AREA "EASY TO SEE"

$$\text{AREA} = \frac{1}{2} \times 1 \times \sqrt{3}$$

$$\text{AREA} = \frac{\sqrt{3}}{2} \text{ SQUARE UNITS}$$

Copyright © Save My Exams. All Rights Reserved
save my exams